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Bayes' Decision Theory

(plus Discriminant Functions and PDF Estimation)

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Agenda

1 Bayes' Classification Rule

2 Discriminant Functions

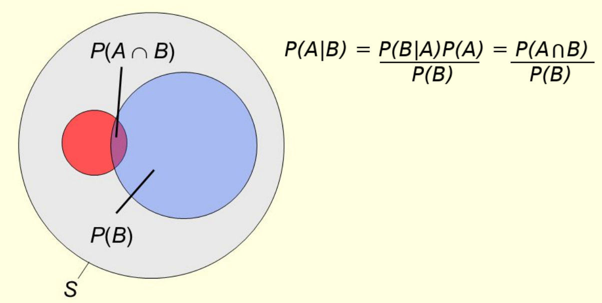
3 Probability Density Function Estimation

Bayes' Rule/Theorem

An important result from Statistical Inference

Let A and B be two arbitrary events:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \tag{1}$$



Bayes for Classification

Let $\mathbf{x} = [x_1, \ldots, x_n]$ be the representation of patterns, for which we want to infer the class in $\Omega = \{\omega_1, \ldots, \omega_c\}$

$$(\mathbf{x}, \omega_j) \Leftrightarrow \arg\max_{\omega_j \in \Omega} P(\omega_j|\mathbf{x}), \tag{2}$$

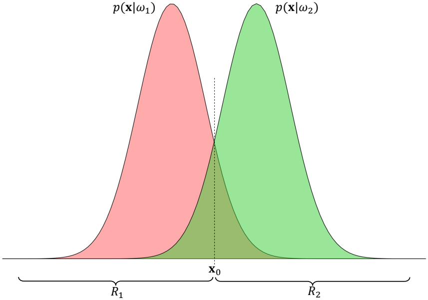
Bayes' Rule allows the inference of $P(\omega_j|\mathbf{x})$:

$$P(\omega_j|\mathbf{x}) = \frac{p(\mathbf{x}|\omega_j)P(\omega_j)}{p(\mathbf{x})}, \tag{3}$$

- $P(\omega_j|\mathbf{x})$ — posterior probability
- $p(\mathbf{x}|\omega_j)$ — likelihood function of ω_j
- $P(\omega_j)$ — prior probability
- $p(\mathbf{x}) = \sum_{k=1}^{c} p(\mathbf{x}|\omega_j)P(\omega_j)$ — evidence probability

Error Notion in Bayes' Rule

- Let classification be between ω_1 and ω_2, equally probable, with $\mathcal{X} \equiv \mathbb{R}$
- $\mathcal{X}$ is divided into the regions R_1 and R_2
- R_1 includes the values of $\mathbf{x}$ such that $P(\omega_1|\mathbf{x}) > P(\omega_2|\mathbf{x})$
- R_2 includes the values such that $P(\omega_2|\mathbf{x}) > P(\omega_1|\mathbf{x})$
- The decision that ensures minimum error occurs at $\mathbf{x}_0$

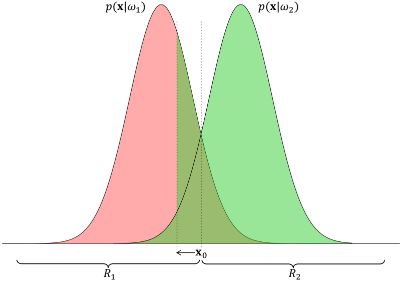


Error Notion in Bayes' Rule (cont.)

The probability of classification error can be expressed as:

$$P_{\text{error}} = P(\mathbf{x} \in R_2, \omega_1) + P(\mathbf{x} \in R_1, \omega_2) = \frac{1}{2}\left[\int_{R_2} p(\mathbf{x}|\omega_1)\,d\mathbf{x} + \int_{R_1} p(\mathbf{x}|\omega_2)\,d\mathbf{x}\right] \tag{4}$$

where $P(\mathbf{x} \in R_2, \omega_1)$ quantifies the probability of pattern $\mathbf{x}$ belonging to region R_2, although its actual class is ω_1.



Changes in the position of $\mathbf{x}_0$ increase the probability of error!

MAP and ML Classification

- $P(\omega_i|\mathbf{x}) > P(\omega_j|\mathbf{x})$, for $i \neq j = 1, \ldots, c$, implies the association of $\mathbf{x}$ to class ω_i
- Equivalently: $\dfrac{p(\mathbf{x}|\omega_i)P(\omega_i)}{p(\mathbf{x})} > \dfrac{p(\mathbf{x}|\omega_j)P(\omega_j)}{p(\mathbf{x})}$ implies the same association
- $p(\mathbf{x})$ does not interfere with class selection...

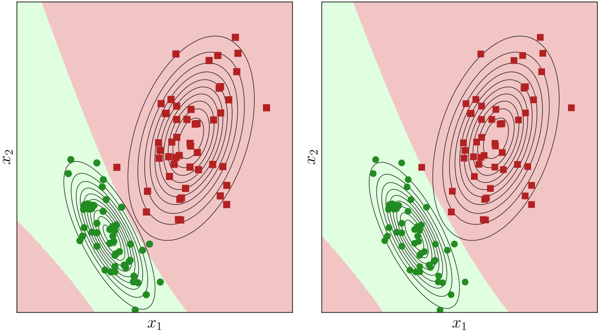
MAP — Maximum a Posteriori Model

$$(\mathbf{x}, \omega_j) \Leftrightarrow \arg\max_{\omega_j \in \Omega} p(\mathbf{x}|\omega_j)P(\omega_j) \tag{5}$$

ML — Maximum Likelihood Model

$$(\mathbf{x}, \omega_j) \Leftrightarrow \arg\max_{\omega_j \in \Omega} p(\mathbf{x}|\omega_j) \tag{6}$$

MAP and ML Classification (cont.)



- MAP (left) — ML (right)
- ω_1 — green circles
- ω_2 — red square
- $P(\omega_1) = 0.05$ and $P(\omega_2) = 0.95$ were assumed — this reduced the decision region favorable to ω_1

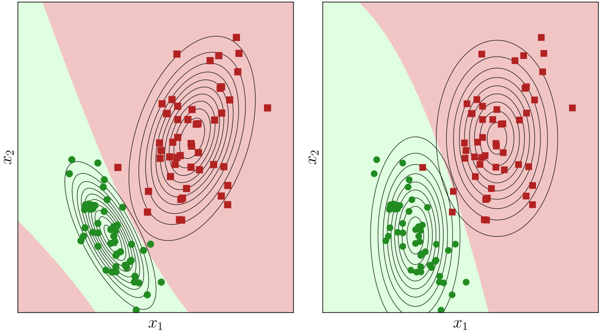
Naive Bayes Classification

- Given that $\mathcal{X}$ is an n-dimensional space, $\mathbf{x}$ becomes n random events $x_1, \ldots, x_n$
- These variables may be independent (though usually they are not)
- Models that do not account for the correlation structure among such variables tend to be fragile
- A naive model assumes $x_1, \ldots, x_n$ are mutually independent

If they are independent, then (by using the "factorization of the conditional joint density" property):

$$
\begin{aligned}
p(\mathbf{x}|\omega_j) &= p(x_1, \ldots, x_n|\omega_j) = p(x_1|x_2, \ldots, x_n, \omega_j)p(x_2, \ldots, x_n|\omega_j) = \\
&= p(x_1|x_2, \ldots, x_n, \omega_j)p(x_2|x_3, \ldots, x_n, \omega_j) \cdots p(x_{n-1}|x_n, \omega_j)p(x_n|\omega_j) = \qquad (7) \\
&= p(x_1|\omega_j)p(x_2|\omega_j) \cdots p(x_n|\omega_j); \quad j = 1, \ldots, c.
\end{aligned}
$$

Naive Bayes (cont.)



- ML (left) — Naive Bayes (right)
- It is easy to see that the decision surface of the Naive Bayes model diverges from the ML model
- The observed effect results from the absence of correlation between attributes x_1 and x_2

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Discriminant Functions and Decision Surfaces

- We have seen the relationship between regions in the feature space, classification error minimization, and the maximization of the posterior probability
- We can define the decision rule as:

$$(\mathbf{x}, \omega_i) \Leftrightarrow P(\omega_i|\mathbf{x}) - P(\omega_j|\mathbf{x}) > 0; \quad i \neq j; \quad j = 1, \ldots, c \tag{8}$$

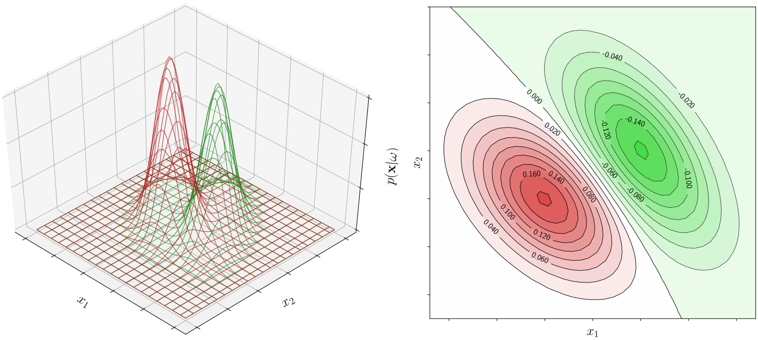
- The geometric locus where $P(\omega_i|\mathbf{x}) - P(\omega_j|\mathbf{x}) = 0$ defines a decision surface

To distinguish between decision rules and regions in the feature space, we reinterpret such rules as *discriminant functions*:

$$g_i(\mathbf{x}) = f(P(\omega_i|\mathbf{x})), \tag{9}$$

where $f(\cdot)$ is a monotonic (increasing or decreasing) function

Discriminant Functions and Decision Surfaces



- The $p(\mathbf{x}|\omega_1)$ (red) and $p(\mathbf{x}|\omega_2)$ (green) probability distributions (left) and the difference $p(\mathbf{x}|\omega_1) - p(\mathbf{x}|\omega_2)$ (right)

Minimum Distance Classification (special cases)

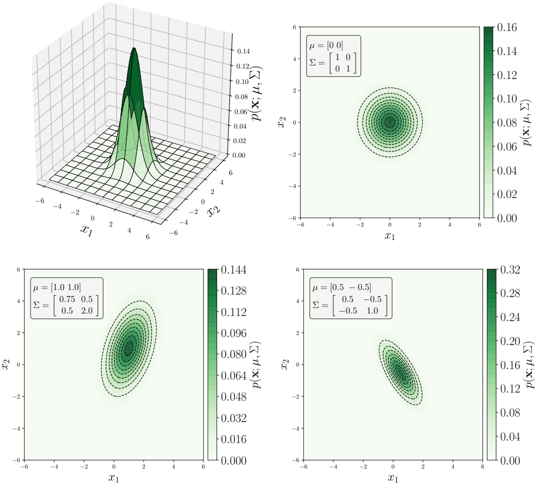
- We saw that it is possible to reinterpret models based on Bayes' rule as discriminant functions.
- Among the three classification models presented, we focus on MAP and assume that the likelihood functions $p(\mathbf{x}|\omega_i)$ follow a Multivariate Gaussian distribution:

$$\mathbf{x} \sim \mathcal{N}(\mu_i, \Sigma_i); \quad \mathbf{x} \in \mathbb{R}^n$$

$$p(\mathbf{x}|\omega_i) = \frac{1}{(2\pi)^{\frac{n}{2}} \, |\Sigma|^{\frac{1}{2}}} e^{-\frac{1}{2}(\mathbf{x}-\mu)^T \Sigma^{-1} (\mathbf{x}-\mu)} \tag{10}$$

- The mean vector and covariance matrix, computed from observed samples in class ω_i, are estimators for parameters μ_i and Σ_i
- To define discriminant functions, we use $f(x) = \ln(x)$ as a monotonically increasing function

Quick Addendum – Multivariate Gaussian – $p(\mathbf{x}|\omega_i) = \frac{1}{(2\pi)^{\frac{n}{2}}|\Sigma|^{\frac{1}{2}}} e^{-\frac{1}{2}(\mathbf{x}-\mu)^T\Sigma^{-1}(\mathbf{x}-\mu)}$



Minimum Distance Classifier...

$$g_i(\mathbf{x}) = \ln\left(p(\mathbf{x}|\omega_i)P(\omega_i)\right) = \ln\left(p(\mathbf{x}|\omega_i)\right) + \ln P(\omega_i)$$

$$= \ln\left(\frac{1}{\sqrt{2\pi\left|\Sigma_i^{-1}\right|}}\right) - \frac{1}{2}\left(\mathbf{x}-\mu_i\right)^T \Sigma_i^{-1}\left(\mathbf{x}-\mu_i\right) + \ln P(\omega_i)$$

$$= -\frac{1}{2}\mathbf{x}^T\Sigma_i^{-1}\mathbf{x} + \frac{1}{2}\mathbf{x}^T\Sigma_i^{-1}\mu_i + \frac{1}{2}\mu_i^T\Sigma_i^{-1}\mathbf{x} - \frac{1}{2}\mu_i^T\Sigma_i^{-1}\mu_i - \frac{1}{2}\ln 2\pi - \frac{1}{2}\ln\left|\Sigma_i^{-1}\right| + \ln P(\omega_i). \quad (11)$$

Since the terms $-\frac{1}{2}\mu_i^T\Sigma_i^{-1}\mu_i - \frac{1}{2}\ln(2\pi)^n - \frac{1}{2}\ln|\Sigma_i| + \ln P(\omega_i)$ are independent of $\mathbf{x}$, we can express them as a constant D_i related to the behavior of ω_i. Therefore, the previous equation is simplified to:

$$g_i(\mathbf{x}) = -\frac{1}{2}\mathbf{x}^T\Sigma_i^{-1}\mathbf{x} + \frac{1}{2}\mathbf{x}^T\Sigma_i^{-1}\mu_i + \frac{1}{2}\mu_i^T\Sigma_i^{-1}\mathbf{x} - D_i \quad (12)$$

Mahalanobis Minimum Distance Classifier

- Assuming $\mathbf{x} \sim \mathcal{N}(\mu_i, \Sigma)$ — class with identical variances/covariances
- Under these conditions:

$$g_i(\mathbf{x}) = -\frac{1}{2}\mathbf{x}^T\Sigma^{-1}\mathbf{x}+\frac{1}{2}\mathbf{x}^T\Sigma^{-1}\mu_i+\frac{1}{2}\mu_i^T\Sigma^{-1}\mathbf{x}-\frac{1}{2}\mu_i^T\Sigma^{-1}\mu_i-\frac{1}{2}\ln 2\pi-\frac{1}{2}\ln|\Sigma|+\ln P(\omega_i). \tag{13}$$

- Independently of $\mathbf{x}$, the terms $-\frac{1}{2}\ln 2\pi - \frac{1}{2}\ln|\Sigma| + \ln P(\omega_i)$ are constants
- Assuming equiprobable classes $\rightarrow \ln P(\omega_i) = \ln P(\omega_j), \forall i, j$
- $\mathbf{x}^T\Sigma^{-1}\mathbf{x}$ is invariant to class change, hence irrelevant to the classification process.

These observations lead to:

$$g_i(\mathbf{x}) = -\frac{1}{2}\left((\mathbf{x}-\mu_i)^T\Sigma^{-1}(\mathbf{x}-\mu_i)\right)^{\frac{1}{2}} \tag{14}$$

The considerations made about variances and covariances define the discriminant function obtained by the Mahalanobis minimum distance classifier, whose associated decision rule is:

$$\begin{aligned}(\mathbf{x}, \omega_i) \Leftrightarrow g_i(\mathbf{x}) > g_j(\mathbf{x});\; i \neq j;\; j = 1, \ldots, c \\ \equiv \mathrm{dm}_i(\mathbf{x}) < \mathrm{dm}_j(\mathbf{x});\; i \neq j;\; j = 1, \ldots, c,\end{aligned} \tag{15}$$

where $\mathrm{dm}_i(\mathbf{x}) = -g_i(\mathbf{x})$.

Euclidean Minimum Distance Classifier

- Assuming Σ is diagonal (zero covariances — independence among $\mathbf{x}$ components) and identical variances: $\Sigma = \sigma^2 \mathbf{I}$
- Consequently:

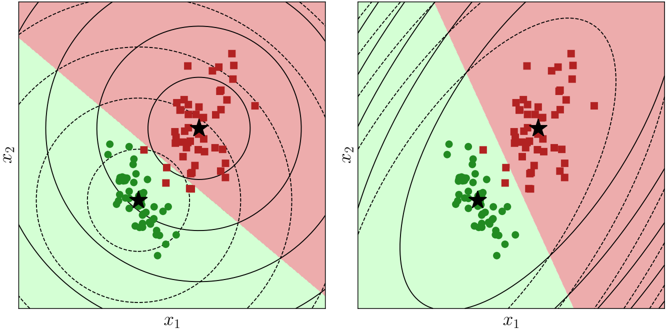
$$g_i(\mathbf{x}) = -\frac{1}{2}\left((\mathbf{x}-\mu_i)^T\left(\sigma^2\mathbf{I}\right)^{-1}(\mathbf{x}-\mu_i)\right)^{\frac{1}{2}}$$

$$= -\frac{1}{2\sigma^2}\left((\mathbf{x}-\mu_i)^T(\mathbf{x}-\mu_i)\right)^{\frac{1}{2}} = \tag{16}$$

$$= -\frac{1}{2\sigma^2}\|\mathbf{x}-\mu_i\|.$$

- The constant $\frac{1}{2\sigma^2}$ does not vary with the class and can be omitted.
- The resulting expression reveals that classification is guided by the Euclidean distance $\mathrm{de}_i(\mathbf{x}) = -g_i(\mathbf{x})$

Mahalanobis and Euclidean Minimum Distance Classifier



Effects of classification using Euclidean minimum distance (right) and Mahalanobis (left). The contour lines, colored according to the corresponding class, indicate the distances from the class means, which are marked with a star.

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Probability Density Function Estimation

- If it is possible to employ a specific distribution model, fitting this model to the particular problem is done through the selection of parameters.
- Among the different techniques for obtaining such parameters, Maximum Likelihood Estimation is a common alternative.
- However, in some cases, the use of known distributions implies unrealistic assumptions or violations regarding the data.
- In such circumstances, mixture models between probability distributions can be used.
- There are also non-parametric methods for estimating probability distributions — these are independent of assumptions about models or parameters.

Gaussian Mixture Model (+EM)

- A mixture model expresses a distribution $p(\mathbf{x})$ as a linear combination of probability density functions:

$$p(\mathbf{x}) = \sum_{j=1}^{k} \lambda_j p(\mathbf{x}|\theta_j); \tag{17}$$

$$\text{with } \sum_{j=1}^{k} \lambda_j = 1; \quad \int_{\mathbf{x}\in\mathcal{X}} p(\mathbf{x}|\theta_j) d\mathbf{x} = 1,$$

  where θ_j is a tuple of parameters of the j-th component.
- We assume $p_j(\mathbf{x}|y_j, \theta_j)$ are multivariate Gaussians, allowing broader applicability to multivariate data — this assumption defines the model as a Gaussian Mixture Model (GMM).
- Based on the dataset $\mathcal{D}$ and a parameter configuration $\Psi = (\lambda_1, \ldots, \lambda_k, \theta_1, \ldots, \theta_k)$, we can compute the membership of each $\mathbf{x}_i$ with respect to component j as:

$$w_{ij} = p(y_j = 1|\mathbf{x}_i, \Psi) = \frac{\lambda_j p_j(\mathbf{x}_i|y_j, \theta_j)}{\sum_{l=1}^{k} \lambda_l p_l(\mathbf{x}_i|y_l, \theta_l)} \tag{18}$$

Gaussian Mixture Model (+EM) (cont.)

- The adjustment is done through two iterative steps: one based on Expectation (E) of the influence of each component in the mixture, and another on Maximization (M) of the probability by updating parameters.
- **E-step:** Starting from an initial configuration (random or not) for $\Psi = (\lambda_j, \theta_j : j = 1, \ldots, k)$, compute membership values w_{ij} for $i = 1, \ldots, m$ and $j = 1, \ldots, k$.
- **M-step:** With the membership values in hand, update the mixture parameters using:

$$\lambda_j^{(new)} = \frac{m_j}{m}; \quad m_j = \sum_{i=1}^{m} w_{ij}; \quad j = 1 \ldots, k; \tag{19}$$

$$\mu_j^{(new)} = \frac{1}{m_j} \sum_{i=1}^{m} w_{ij} \mathbf{x}_i; \quad j = 1 \ldots, k; \tag{20}$$

$$\Sigma_j^{(new)} = \frac{1}{m_j} \sum_{i=1}^{m} w_{ij} (\mathbf{x}_i - \mu_j^{(new)})^T (\mathbf{x}_i - \mu_j^{(new)}); \quad j = 1 \ldots, k; \tag{21}$$

Gaussian Mixture Model (+EM) (cont.)

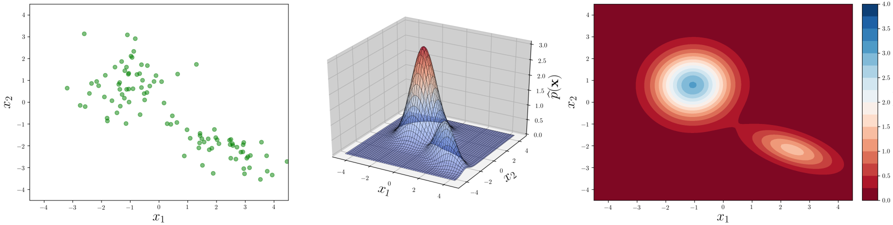
- Once the updates $\lambda_j^{(new)}$, $\mu_j^{(new)}$ and $\Sigma_j^{(new)}$ are obtained, with $j = 1 \ldots, k$, convergence can be checked using the log-likelihood function defined by:

$$\log \mathcal{L}(\Psi^{(new)}) = \sum_{i=1}^{m} \left(\log \left(\sum_{j=1}^{k} \lambda_j p_j(\mathbf{x}_i | z_j, \theta_j) \right) \right), \tag{22}$$

  where $p_j(\mathbf{x}_i | z_j, \theta_j)$ is the Gaussian distribution of the j-th component.

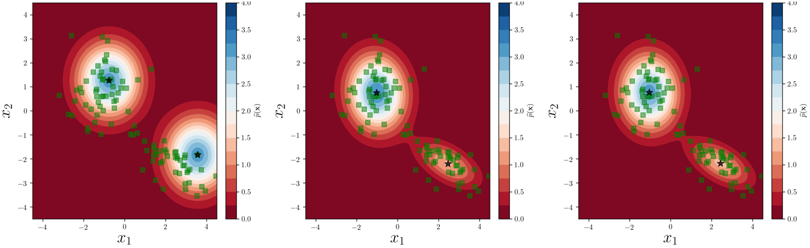
- Thus, as long as convergence is not verified by $\left| \log \mathcal{L}(\Psi^{(new)}) - \log \mathcal{L}(\Psi) \right| < \epsilon$, for some $\epsilon \in \mathbb{R}_+$ as small as desired, the process returns to the **M-step**, where the membership values w_{ij} for $i = 1, \ldots, m$ and $j = 1, \ldots, k$ are recalculated, assuming the parameter configuration $\Psi := \Psi^{(new)}$.

Gaussian Mixture Model (+EM)



Random dataset (left) and fitted models (center and right)

Gaussian Mixture Model (+EM)



Considering as convergence criterion a change smaller than 0.1 in the log-likelihood function between two successive iterations, convergence is reached at the end of the eighth iteration.

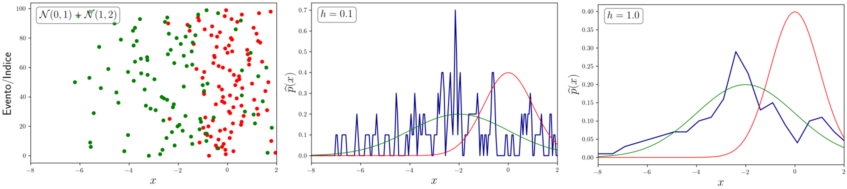
Iterations: 0 (left) – 2 (center) – 8 (right)

Histogram Method

- A simple way to approximate the probability distribution of a given dataset $\mathcal{D} = \{x_i \in \mathbb{R} : i = 1, \ldots, m\}$
- The value range of $\mathcal{D}$ is partitioned into disjoint subintervals of width h, enabling the estimation:

$$\widehat{p}(x) = \frac{1}{h}\frac{q(x)}{m}, \tag{23}$$

where $q(x)$ counts the number of observations in $\mathcal{D}$ that fall within the same subinterval as x



Parzen Windows

- Non-parametric approximation of $\mathcal{D} = \{\mathbf{x}_i \in \mathbb{R}^n : i = 1, \ldots, m\}$
- Density is estimated using hypercubes of side h (and hypervolume h^n), through:

$$\phi(\mathbf{x}_i) = \begin{cases} 1; & \text{if } |x_{ij}| < 1/2;\ j = 1, \ldots, n \\ 0; & \text{otherwise} \end{cases} \tag{24}$$

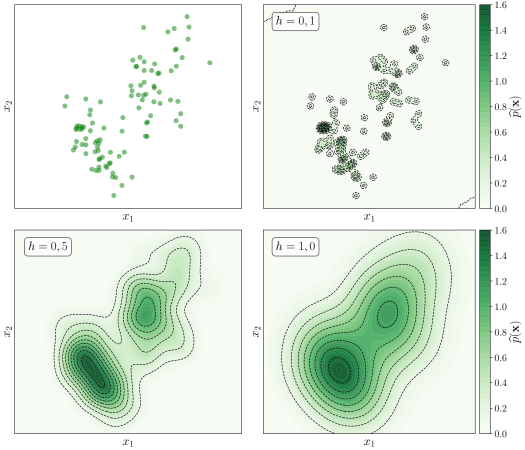
- Replacing $\mathbf{x}_i$ with $\left(\dfrac{\mathbf{x}_i - \mathbf{x}}{h}\right)$ is equivalent to a translation that moves the hypercube from the origin to the point $\mathbf{x}$, followed by a scaling transformation, making it analogous to a hypercube of side h.

$$\widehat{p}(\mathbf{x}) = \frac{1}{h^n}\frac{1}{m}\sum_{i=1}^{m} \phi\left(\frac{\mathbf{x}_i - \mathbf{x}}{h}\right) \tag{25}$$

- The above $\phi(\cdot)$ introduces discontinuities — to smooth the approximation, we use:

$$\phi(\mathbf{z}) = \frac{1}{\sqrt{2\pi}} e^{-\left(\frac{\mathbf{z}^T\mathbf{z}}{2}\right)} \tag{26}$$

Parzen Windows



Application of the Parzen Window method to a random dataset generated by the sum of two multivariate Gaussians, considering different values of h.

k-Nearest Neighbors

- For Parzen Windows — the estimation of $p(\mathbf{x})$ is based on the number of observations in $\mathcal{D}$ that fall within a hypercube of fixed volume
- Conversely, we can fix a number $k \in \mathbb{N}$ of observations, and determine the volume of the smallest hypersphere that contains k observations
- Low-density regions — large volumes; high-density — small volumes

The expression for such estimation is given by:

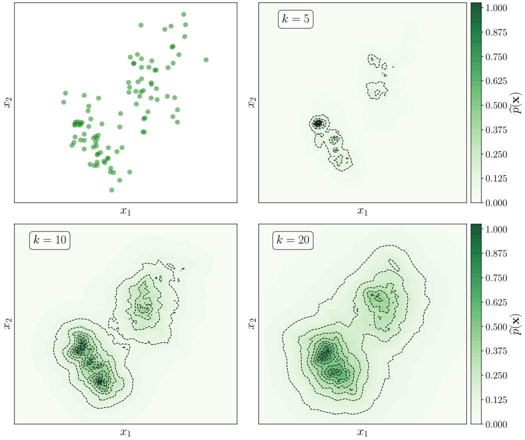
$$\widehat{p}(\mathbf{x}) = \frac{k}{mV_k(\mathbf{x})}, \tag{27}$$

where m is the cardinality of $\mathcal{D}$ and $V_k(\mathbf{x})$ corresponds to the volume of the smallest possible hypersphere centered at $\mathbf{x}$ that contains k observations from $\mathcal{D}$

$$V_k(\mathbf{x}) = V_0\rho^n; \quad \text{with } V_0 = \begin{cases} \frac{\pi^{n/2}}{(n/2)!}, & \text{if } n \text{ is even} \\ 2\pi^{(n-1)/2}\frac{((n-1)/2)!}{n!}, & \text{if } n \text{ is odd} \end{cases} \tag{28}$$

where ρ is the k-th smallest distance observed between $\mathbf{x}$ and the elements of $\mathcal{D}$

K-Nearest Neighbors



Application of the k-Nearest Neighbors method to a random dataset generated by the sum of two multivariate Gaussians, considering different values of k.

Classification via k-Nearest Neighbors

- Consider a dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i) : i = 1, \ldots, m\}$, in which $\mathbf{x}_i$ is associated with a class from $\Omega = \{\omega_1, \omega_2\}$ through class indicators $y_i \in \mathcal{Y} = \{1, 2\}$
- It is possible to estimate class-conditional probability densities:

$$p(\mathbf{x}|\omega_j, k) = \frac{k}{m_j V_0 \rho_{kj}(\mathbf{x})}, \tag{29}$$

  where $\rho_{kj}(\mathbf{x})$ is the k-th smallest distance observed between $\mathbf{x}$ and each of the m_j observations $\mathbf{x}_i$ in $\mathcal{D}$ that belong to class ω_j
- Based on the ML model, the classification rule is:

$$\begin{aligned} &(\mathbf{x}, \omega_j) \Leftrightarrow p(\mathbf{x}|\omega_j, k) > p(\mathbf{x}|\omega_\ell, k) \Leftrightarrow \\ &\Leftrightarrow \frac{k}{m_j V_0 \rho_{kj}(\mathbf{x})} > \frac{k}{m_\ell V_0 \rho_{k\ell}(\mathbf{x})} \Leftrightarrow m_\ell \rho_{k\ell}(\mathbf{x}) > m_j \rho_{kj}(\mathbf{x}) \end{aligned} \tag{30}$$

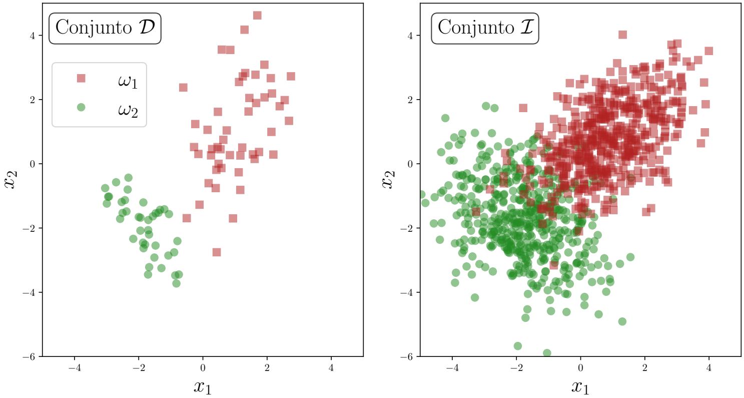
- Interpreting this in terms of distances $\mathrm{dk}_j(\mathbf{x}) = m_j \rho_{kj}(\mathbf{x})$, we have:

$$(\mathbf{x}, \omega_j) \Leftrightarrow \arg\min_j \mathrm{dk}_j(\mathbf{x}), \tag{31}$$

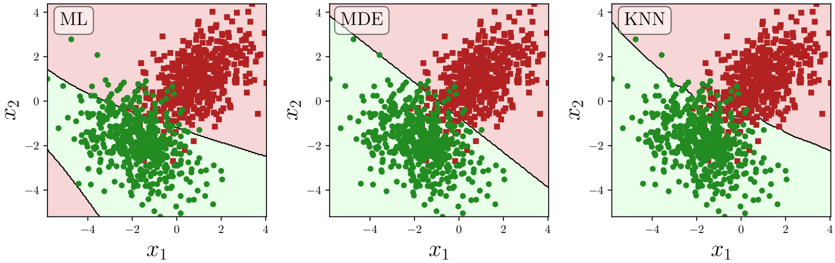
  a decision rule that characterizes the k-Nearest Neighbors (KNN) classifier

Computational Experiment

- Apply the ML, EMD, and KNN ($k = 5$) models
- Train on $\mathcal{D}$ (left) and classify $\mathcal{I}$ (right) — both are simulated
- Compute prediction accuracy



Computational Experiment



Accuracy: ML (left) – 84.2%; EMD (center) – 92.5%; KNN (right) – 93.1%

Bayes' Decision Theory
(plus Discriminant Functions and PDF Estimation)

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